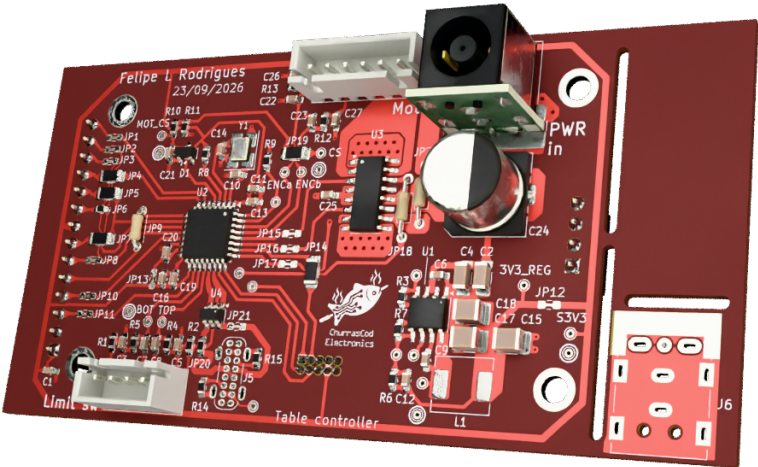
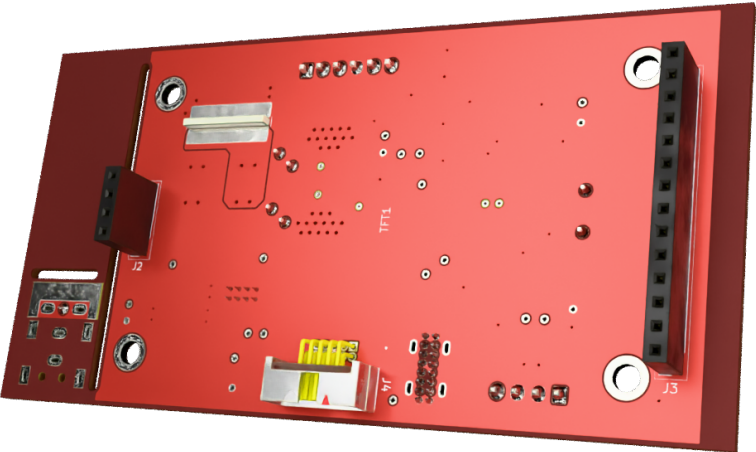


# Desk controller assembly drawing

Top view



Bottom view

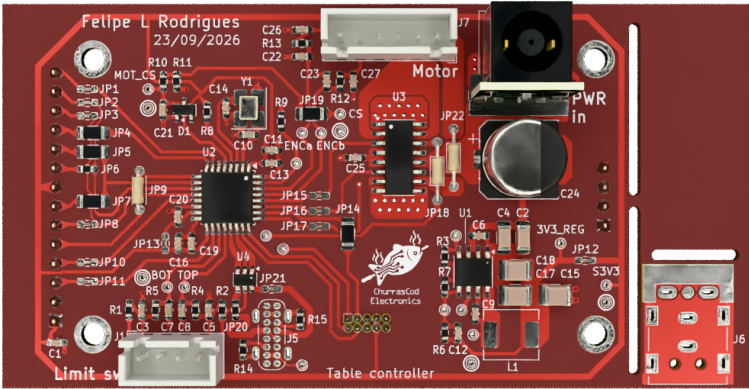
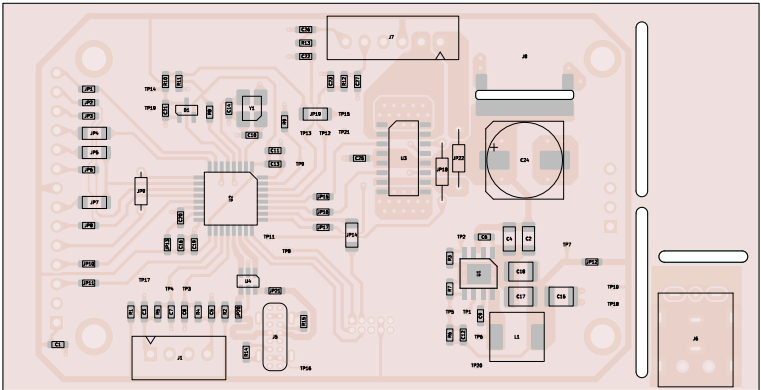


[Interactive BOM](#)

|                                                                                       |                        |                      |                          |            |
|---------------------------------------------------------------------------------------|------------------------|----------------------|--------------------------|------------|
|  | <b>Desk controller</b> |                      | Sheet: Assembly overview |            |
|                                                                                       | <b>Notes</b>           |                      | File: ktest.kicad_pcb    |            |
|                                                                                       | c1                     |                      | Rev: 0.0.2               | Drwn: FLR  |
|                                                                                       | c2                     |                      | Mat: Concept             | Rev'd: FLR |
|                                                                                       | c3                     |                      | Date: 06/10/2026         | RLs'd: FLR |
|                                                                                       |                        | Dwg: NA              | Size: A4                 |            |
|                                                                                       |                        | Generated with KiCad |                          | Page: 1/7  |

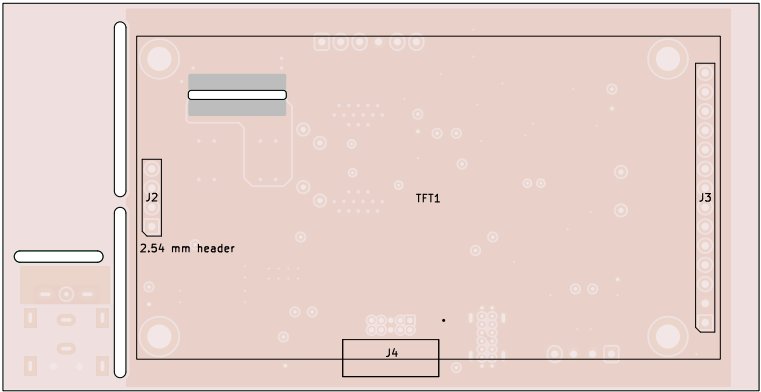
Top assembly (Scale 1:1)

| Type  | Fitted | No fit |
|-------|--------|--------|
| SMD   | 58     | 0      |
| THT   | 6      | 0      |
| Total | 64     | 0      |

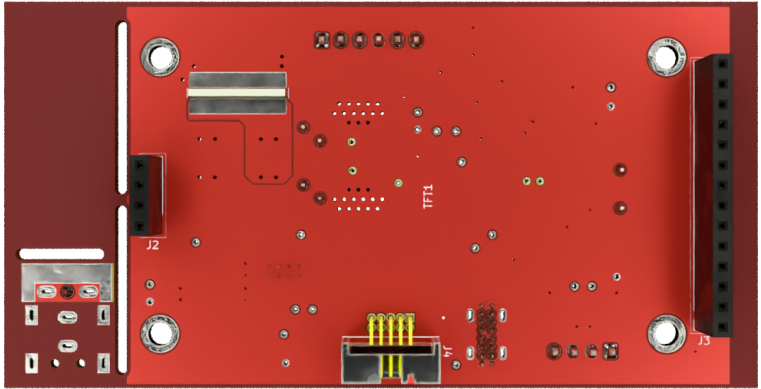


| Desk controller |    | Sheet: Top assembly (Scale 1:1) |            |
|-----------------|----|---------------------------------|------------|
| Notes           |    | File: ktest.kicad_pcb           |            |
|                 | c1 | Rev: 0.0.2                      | Drwn: FLR  |
|                 | c2 | Mat: Concept                    | Rev'd: FLR |
|                 | c3 | Date: 06/10/2026                | Rs'd: FLR  |
|                 | c4 | Dwg: NA                         | Size: A4   |
|                 |    | Generated with KiCad            | Page: 2/7  |

Bottom assembly (Scale 1:1)



| Type  | Fitted | Not fitted |
|-------|--------|------------|
| SMD   | 0      | 0          |
| THT   | 4      | 0          |
| Total | 4      | 0          |



|                        |  |                                                             |                                       |
|------------------------|--|-------------------------------------------------------------|---------------------------------------|
| <b>Desk controller</b> |  | Sheet: Bottom assembly (Scale 1:1)<br>File: ktest.kicad_pcb |                                       |
| <b>Notes</b>           |  | Rev: 0.0.2<br>Mat: Concept<br>Date: 06/10/2026              | Drwn: FLR<br>Rev'd: FLR<br>Rls'd: FLR |
| c1<br>c2<br>c3<br>c4   |  | Dwg: NA<br>Generated with KiCad                             | Size: A4<br>Page: 3/7                 |

# Top BOM - fitted (1 of 3)

| References                  | Qty | Value          | Footprint                                       | MPN | Manufacturer | Status |
|-----------------------------|-----|----------------|-------------------------------------------------|-----|--------------|--------|
| C1 C6 C9 C13 C19<br>C20 C25 | 7   | 100n           | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C3 C5 C26 C27               | 4   | 10n            | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C7 C8 C21 C22 C23           | 5   | 1n             | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C11 C16                     | 2   | 1u             | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C24                         | 1   | 220u           | CP_Elec_10x10                                   |     |              |        |
| C10 C14                     | 2   | 22p            | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C15 C17 C18                 | 3   | 22u            | C_1210_3225Metric_Pad1.33x2.70m<br>m_HandSolder |     |              |        |
| C12                         | 1   | 33p            | C_0603_1608Metric_Pad1.08x0.95m<br>m_HandSolder |     |              |        |
| C2 C4                       | 2   | 4.7u           | C_1206_3216Metric_Pad1.33x1.80m<br>m_HandSolder |     |              |        |
| D1                          | 1   | BAV99          | SOT-23                                          |     |              |        |
| J6                          | 1   | DC barrel assy | Dell_barrel_assy                                |     |              |        |
| J8                          | 1   | Dell DC jack   | Dell DC jack                                    |     |              |        |



## Desk controller

### Notes

c1  
c2  
c3  
c4

Sheet: Top BOM - fitted (1 of 3)

File: ktest.kicad\_pcb

Rev: 0.0.2

Mat: Concept

Date: 06/10/2026

Dwg: NA

Generated with KiCad

Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

Page: 4/7

## Top BOM - fitted (2 of 3)

| References               | Qty | Value                           | Footprint                                            | MPN | Manufacturer | Status |
|--------------------------|-----|---------------------------------|------------------------------------------------------|-----|--------------|--------|
| J1                       | 1   | JST B4B XH A                    | JST_XH_B4B-XH-A_1x04_P2.50mm_Ve<br>rtical            |     |              |        |
| J7                       | 1   | JST B6B XH A                    | JST_XH_B6B-XH-A_1x06_P2.50mm_Ve<br>rtical            |     |              |        |
| J5                       | 1   | USB_C_Receptac<br>le_USB2.0_16P | USB_C_socket_16P_vertical                            |     |              |        |
| JP4 JP5 JP7 JP14<br>JP19 | 5   | 0R                              | R_1206_3216Metric_Pad1.30x1.75m<br>m_HandSolder      |     | Generic      |        |
| JP18 JP22                | 2   | 0R                              | R_Axial_DIN0204_L3.6mm_D1.6mm_P<br>7.62mm_Horizontal |     | Generic      |        |
| JP6 JP20                 | 2   | 0R                              | R_0603_1608Metric_Pad0.98x0.95m<br>m_HandSolder      |     | Yageo        |        |
| JP9                      | 1   | 0R                              | R_Axial_DIN0204_L3.6mm_D1.6mm_P<br>5.08mm_Horizontal |     | Generic      |        |
| L1                       | 1   | 10u                             | L_Bourns_SRP6540                                     |     |              |        |
| R1 R2 R7 R8 R9           | 5   | 10k                             | R_0603_1608Metric_Pad0.98x0.95m<br>m_HandSolder      |     |              |        |
| R6                       | 1   | 23.2k                           | R_0603_1608Metric_Pad0.98x0.95m<br>m_HandSolder      |     |              |        |
| R4 R5 R10 R12 R13        | 5   | 2k2                             | R_0603_1608Metric_Pad0.98x0.95m<br>m_HandSolder      |     |              |        |



### Desk controller

#### Notes

c1  
c2  
c3  
c4

Sheet: Top BOM - fitted (2 of 3)

File: ktest.kicad\_pcb

Rev: 0.0.2  
Mat: Concept  
Date: 06/10/2026

Dwg: NA

Generated with KiCad

Drwn: FLR  
Rev'd: FLR  
Rls'd: FLR

Size: A4

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## Top BOM - fitted (3 of 3)

| References | Qty | Value         | Footprint                                      | MPN | Manufacturer | Status |
|------------|-----|---------------|------------------------------------------------|-----|--------------|--------|
| R14 R15    | 2   | 5k1           | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder    |     |              |        |
| R3         | 1   | 64.9k         | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder    |     |              |        |
| R11        | 1   | 820           | R_0603_1608Metric_Pad0.98x0.95mm_HandSolder    |     |              |        |
| U1         | 1   | LMR380x0      | Texas_HSOP-8-1EP_3.9x4.9mm_P1.27mm_ThermalVias |     |              |        |
| U2         | 1   | STM32G431K8Tx | LQFP-32_7x7mm_P0.8mm                           |     |              |        |
| U4         | 1   | USBLC6-2SC6   | SOT-23-6                                       |     |              |        |
| U3         | 1   | VNH7070       | JEITA_SOIC-16_3.9x9.9mm_P1.27mm                |     |              |        |
| Y1         | 1   | 20 MHz        | Crystal_SMD_3225-4Pin_3.2x2.5mm_HandSoldering  |     |              |        |

|                                                                                       |                        |  |                                  |            |
|---------------------------------------------------------------------------------------|------------------------|--|----------------------------------|------------|
|  | <b>Desk controller</b> |  | Sheet: Top BOM - fitted (3 of 3) |            |
|                                                                                       | <b>Notes</b>           |  | File: ktest.kicad_pcb            |            |
|                                                                                       | c1                     |  | Rev: 0.0.2                       | Drwn: FLR  |
|                                                                                       | c2                     |  | Mat: Concept                     | Rev'd: FLR |
|                                                                                       | c3                     |  | Date: 06/10/2026                 | RLs'd: FLR |
|                                                                                       |                        |  | Dwg: NA                          | Size: A4   |
|                                                                                       | c4                     |  | Generated with KiCad             | Page: 6/7  |

Bottom BOM - fitted (1 of 1)

| References | Qty | Value                    | Footprint                                 | MPN | Manufacturer | Status |
|------------|-----|--------------------------|-------------------------------------------|-----|--------------|--------|
| J2         | 1   | 2.54 mm header           | PinSocket_1x04_P2.54mm_Vertical           |     |              |        |
| J3         | 1   | Conn_01x14_Pin           | PinSocket_1x14_P2.54mm_Vertical           |     |              |        |
| J4         | 1   | HX JN1.27-2x5<br>WZ H4.9 | JN1.27-2x5 WZ H4.9                        |     |              |        |
| TFT1       | 1   | TFT_2.4in_320x<br>240    | Display_240x320_ILI9341_2.4in_n<br>o_pads |     |              |        |



|                               |  |                                     |            |
|-------------------------------|--|-------------------------------------|------------|
| Desk controller               |  | Sheet: Bottom BOM - fitted (1 of 1) |            |
| Notes<br>c1<br>c2<br>c3<br>c4 |  | File: ktest.kicad_pcb               |            |
|                               |  | Rev: 0.0.2                          | Drwn: FLR  |
|                               |  | Mat: Concept                        | Rev'd: FLR |
|                               |  | Date: 06/10/2026                    | RLs'd: FLR |
|                               |  | Dwg: NA                             | Size: A4   |
|                               |  | Generated with KiCad                | Page: 7/7  |